Quadratic Functions (2.1) Name ____________________

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $(2, 0)$ # ______ State the vertex of $f(x) = (x-3)^2 + 2$.	ANSWER: $\{-3,\ 3\}$ # ______ Find the x-value that minimizes $f(x) = x^2 - 8x + 1$.
ANSWER: $x = 3$ # ______ State the y-intercept of $f(x) = x^2 - 4x + 7$.	ANSWER: $\{-2, -1, 1, 2\}$ # ______ Does the parabola $f(x) = -2x^2 + 3x$ open up or down?

ANSWER: $(3, 2)$ # ________ State the axis of symmetry of $f(x) = x^2 - 6x + 5$.	ANSWER: 0 # ________ Solve $x^4 - 5x^2 + 4 = 0$.
ANSWER: $(-1, -1)$ # ________ Solve $x^2 - 9 = 0$.	ANSWER: down # ________ Write the vertex of $y = x^2 - 4x + 4$ in vertex form's terms.
ANSWER: $(0, 7)$ # ________ Find the vertex of $f(x) = x^2 + 2x$.	ANSWER: 4 # ________ How many x-intercepts does $f(x) = x^2 + 4$ have?